

AccessControlPro - User Manual

Version 3.4 | Gym & Fitness Club Management System with Access Control, POS, and Financial Management

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1. System Overview

AccessControlPro is a comprehensive gym management system that includes:

- **Main App (WPF)** - Daily gym operations: player management, access control, events, finance
- **Admin Panel** - System administration: users, products, suppliers, reports, backup
- **POS Terminal** - Point-of-sale: product sales, card balance, barcode scanning

System Requirements

Component	Requirement
Operating System	Windows 10 / 11 (64-bit)
Database	SQL Server 2019+ or SQL Server Express

RAM	Minimum 4 GB
Storage	500 MB for application + database
.NET Runtime	Not required (self-contained)
Access Control Hardware	FK-compatible controllers (optional)

Key Features

- Player subscription management with card-based access control
 - Real-time device monitoring and event logging
 - QR code passes for daily/temporary visitors
 - Point-of-sale with inventory management
 - Financial tracking (income, expenses, outstanding balances)
 - Bilingual interface (English / Arabic)
 - Role-based user permissions
 - Data encryption and security hardening
 - Multi-monitor support for entrance display
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2. Installation & Setup

First-Time Installation

1. Copy the **WPF** folder to the target PC (e.g., C:\AccessControlPro)
2. Run `AccessControlPro.WPF.exe`
3. The **Setup Wizard** will launch automatically on first run

Setup Wizard Steps

Step 1: Welcome

- Welcome screen with application information
- Developer logo upload (optional)

Step 2: Database Connection

- **Server:** Enter SQL Server name (e.g., `localhost` , `.\SQLEXPRESS` , or remote IP)
- **Database:** Name for the database (default: `AccessControlPro`)
- **Authentication:** SQL Server credentials (User ID + Password) or Windows Authentication
- Click **Test Connection** to verify before proceeding

Step 3: Gym Information

- **Gym Name:** Your gym/club name (appears in reports and receipts)
- **Currency:** Currency symbol for financial displays (e.g., IQD, USD, EUR)
- Gym logo upload (optional, displayed in reports)

Step 4: Admin Account

- **Username:** Administrator login name
- **Password:** Must meet password policy (minimum 8 characters, mixed case, number)
- **Email:** Admin email for recovery

Step 5: Finish

- Review configuration summary
- Option to create desktop shortcuts for all 3 apps
- Click **Finish** to complete setup

Installing Additional Apps

For the **Admin Panel** and **POS Terminal**:

1. Copy the `Admin` and/or `POS` folders to the same PC or other PCs
 2. Edit `appsettings.json` in each folder to match the same database connection string
 3. Run the respective `.exe` file
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3. Login & Authentication

Logging In

1. Launch the application
2. Enter your **Username** and **Password**
3. Click **Login** or press **Enter**
4. Check **Remember Me** to save your username for next login

Security Protections

- **Brute force protection:** Account locks after 5 failed attempts (15-minute cooldown)
- **Password policy:** Minimum 8 characters with complexity requirements
- **Session tracking:** All actions are logged with the current user

Changing Password

1. Click the **key icon** in the bottom sidebar
2. Enter your current password
3. Enter and confirm your new password
4. Click **Change Password**

Language Toggle

- Click the **globe icon** in the bottom sidebar to switch between English and Arabic
 - The interface updates immediately without restart
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4. Main Application

4.1 Dashboard

The dashboard provides a real-time overview of your gym operations:

Card	Description
Total Devices	Number of access control devices registered
Online Devices	Devices currently connected and responding
Total Doors	Number of doors configured across all devices
Today's Events	Access events recorded today
Active Alarms	Current alarm conditions

- **Recent Events:** Shows the last 20 access events with door name, card number, event type, and timestamp
- **Door Status:** Shows all doors with their current lock/unlock state
- Click **Refresh** to update all statistics

4.2 Device Management

Manage your access control hardware devices.

Adding a Device

1. Click **Add Device**
2. Enter device details:

- **Name:** Friendly name (e.g., "Main Entrance")
- **IP Address:** Device network IP
- **Port:** Communication port (default: 8000)
- **Username/Password:** Device credentials

3. Click **Save**

Network Discovery

1. Click **Search Network** to auto-discover devices on your local network
2. Discovered devices will appear in the list
3. Select and add devices to your system

Connecting to a Device

1. Select a device from the list
2. Click **Connect** to establish communication
3. The status indicator will show green (online) or red (offline)

Downloading Event Logs

1. Select an online device
2. Click **Download Logs**
3. The system verifies device connectivity first
4. Select the date period for log download
5. Events are downloaded and saved to the database
6. A summary dialog shows the number of records imported

4.3 Door Management

Configure doors linked to your access control devices.

Adding a Door

1. Click **Add Door**
2. Select the **parent device**
3. Enter door details:
 - **Name:** Door name (e.g., "Front Door", "VIP Room")
 - **Door Number:** Physical door number on the device (1-4)
4. Click **Save**

4.4 Player Management

The core of daily operations - managing gym members and their access cards.

Adding a New Player

1. Click **Add Player**
2. Fill in player details:
 - **Full Name (EN):** English name
 - **Full Name (AR):** Arabic name
 - **Phone:** Contact number
 - **Card No:** Member card/ID number
 - **Subscription Type:** Select from configured plans
 - **Duration:** Months or custom days
 - **Fee / Amount Paid:** Subscription cost and payment received
 - **Photo:** Upload player photo (optional)
3. Click **Save**

Assigning an Access Card

1. Select a player from the list
2. Click **Assign Card**

3. Enter card details:
 - **Card Number:** Physical card number
 - **Door Permissions:** Select which doors the card can access
 - **Effective Times:** Access schedule
4. Select **target devices** to sync the card to
5. The system checks device connectivity before syncing
6. Click **Assign** to save and sync

Renewing a Subscription

1. Select a player
2. Click **Renew**
3. Choose:
 - **Subscription Type:** New or same plan
 - **Duration:** Months or custom days
 - **Fee / Amount Paid**
 - **Door Permissions:** Update if needed
4. Select devices to sync updated card validity
5. Click **Renew**
6. A receipt is generated for the transaction

Freezing a Player

1. Select a player
2. Click **Freeze**
3. Enter a **freeze reason** (e.g., "Travel", "Medical")
4. The player's subscription is paused
5. Cards are disabled on hardware devices
6. When unfreezing, the remaining subscription days are automatically added back

Unfreezing a Player

1. Select a frozen player
2. Click **Unfreeze**
3. Confirm the action
4. The subscription end date is extended by the freeze duration
5. Cards are re-enabled on hardware devices

Bulk Operations

1. Click **Bulk Operations**
2. Select operation type:
 - **Freeze All:** Freeze multiple players at once
 - **Unfreeze All:** Unfreeze multiple players
 - **Extend Subscription:** Add days to multiple players
3. Select target group:
 - All active players
 - Expiring in 7 days
 - Expiring in 30 days
 - Currently filtered list
4. Confirm and execute

Filtering Players

Use the filter chips at the top:

- **All:** Show all players
- **Expiring:** Players expiring within 7 days
- **Renewed:** Recently renewed players
- **Frozen:** Currently frozen players

- **Expired:** Past-due subscriptions

Use the **search bar** to find players by name, card number, phone, or subscription type.

4.5 Access Events

View and manage access events from all devices.

Viewing Events

- Events show: timestamp, door name, card number, event type, description
- Use **pagination** to browse through large event lists (100 per page)

Filtering Events

- **By Type:** Card action, Button, Door sensor, Software, Alarm, System
- **By Device:** Select specific device
- **By Period:** Today, Yesterday, This week, Last week, This month, Last month, Last 3/6 months, This year, or Custom range

Downloading Device Logs

1. Select a device from the dropdown
2. Click **Download Logs**
3. Select date period
4. The system downloads events directly from the device hardware

4.6 Finance

Track your gym's financial health.

Overview

- **Total Revenue:** All income in the selected period
- **Total Expenses:** All expenses in the selected period
- **Net Profit:** Revenue minus expenses
- **Unpaid Balances:** Outstanding subscription payments

Recent Transactions

- Shows the most recent transactions with type, category, amount, and date
- Filter by period (Today, This week, This month, etc.)
- Search by description or player name

Outstanding Balances

- Lists players who have unpaid subscription balances
- Shows: Player name, subscription fee, amount paid, remaining balance
- Click **Pay** to record a partial or full payment

4.7 Cash Flow

Manage daily income and expense transactions.

Adding Income

1. Click **Add Income**
2. Select **category** (Subscription, POS Sales, Owner Deposit, Other)
3. Enter **amount** and **description**
4. Select **payment method**
5. Click **Save**

Adding Expense

1. Click **Add Expense**
2. Select **category** (Rent, Electricity, Water, Salaries, Equipment, etc.)
3. Enter **amount** and **description**

4. Click **Save**

4.8 QR Passes

Create temporary QR code passes for daily visitors.

Creating a QR Pass

1. Click **Create Pass**
2. Enter visitor details:
 - **Player Name**: Visitor name
 - **Phone**: Contact number
 - **Fee**: Pass fee amount
 - **Max Uses**: Maximum number of entries (default: 5)
 - **Valid Days**: Number of days the pass is active
 - **Device/Door**: Select which device and door for access
3. Click **Create**
4. A QR code is generated and displayed

QR Code Display

- The QR code can be printed or saved
- Share with the visitor for scanning at the entrance
- The pass code is unique and tamper-proof

Scanning QR Passes

- At the entrance, the **QR Scan Window** accepts scanner input
- The system validates: pass code, expiry date, remaining uses
- Access is granted or denied with visual feedback

4.9 Monitor Display

Display real-time access events on a secondary monitor (e.g., entrance TV).

Setup

1. Connect a secondary monitor/TV to your PC
2. Click **Monitor** in the sidebar
3. The monitor window automatically detects and positions on the secondary display
4. Shows: player photo, name, access status, timestamp

Controls

- **Start/Stop**: Toggle real-time monitoring
- **Projector Mode**: Optimize display for projection
- **Full Screen**: Toggle full-screen mode

4.10 Logs & Audit

View session activity logs.

- All operations are logged: CREATE, UPDATE, DELETE, FREEZE, UNFREEZE, RENEW, ASSIGN_CARD, REMOVE_CARD, SYNC_CARD
- Each log entry includes: timestamp, operation, entity, user
- Logs are stored in `Logs/session.log`
- Session logs are reset on each app restart

4.11 Deleted Records

View and manage soft-deleted players.

Viewing Deleted Records

- Shows all players that were soft-deleted (not permanently removed)

- Displays: player name, deletion date, deleted by, reason

Restoring a Record

1. Select a deleted player
2. Click **Restore**
3. The player is restored to active status

4.12 Data Migration

Import and export data between AccessControlPro instances.

1. Click **Data Migration** in the sidebar
 2. Choose **Import** or **Export**
 3. Select tables to migrate
 4. Follow the wizard to complete the transfer
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5. Admin Panel

The Admin Panel provides system-level management.

5.1 User Management

Manage system users and their permissions.

Creating a User

1. Go to **Users**
2. Click **Add User**
3. Enter: Username, Password, Email
4. Assign **permissions** per application:

Main App Permissions:

Permission	Description
Dashboard	View dashboard
Devices	View devices
DevicesAdd/Edit/Delete/Connect	Device management
Doors	View doors
DoorsAdd/Edit/Delete	Door management
Players	View players
PlayersAdd/Edit/Delete	Player CRUD
PlayersAssignCard/RemoveCard	Card operations
PlayersFreeze/Renew	Subscription operations
PlayersReports	Player reports
Events	View events
Finance/CashFlow	Financial access
FinanceManage/CashFlowManage	Financial modifications
Logs/DeletedRecords	View logs/deleted

Monitor	Monitor display
QrPass	QR pass management

5. Click **Save**

Editing Permissions

1. Select a user
2. Click **Edit**
3. Modify permissions as needed
4. Click **Save**

5.2 Categories & Products

Managing Categories

- Add product categories (Drinks, Supplements, Gear, Accessories, etc.)
- Categories are used in the POS terminal for product organization
- Each category has English and Arabic names

Managing Products

1. Go to **Products**
2. Click **Add Product**
3. Enter:
 - **Name (EN/AR)**: Product name in both languages
 - **Barcode**: Product barcode (optional, must be unique)
 - **Price**: Selling price
 - **Category**: Product category
 - **Stock**: Initial stock quantity
4. Click **Save**

5.3 Suppliers & Purchase Orders

Managing Suppliers

1. Go to **Suppliers**
2. Click **Add Supplier**
3. Enter: Name, Phone, Address, Contact Person
4. Click **Save**

Creating Purchase Orders

1. Go to **Purchase Orders**
2. Click **New Order**
3. Select **Supplier**
4. Add products with quantities and unit costs
5. Set **discount** (if any) and **amount paid**
6. Click **Create**
7. Stock is automatically updated for each product
8. An expense transaction is automatically created

Payment Tracking

- Purchase orders track: Total, Discount, Amount Paid, Status (Paid/Partial/Unpaid)
- Make additional payments by clicking **Pay** on a purchase order

5.4 Reports

Generate business reports with date filtering:

- **Membership Reports**: Active, expired, frozen players
- **Financial Reports**: Revenue, expenses, profit by period

- **Attendance Reports:** Access events by player/door
- **Device Reports:** Device uptime and event counts

5.5 Audit Log

View all security-related events:

- User logins/logouts
- Data modifications (create, update, delete)
- Permission changes
- Card operations
- Device connections
- Filter by date, user, action type

5.6 Backup & Restore

Creating a Backup

1. Go to **Backup**
2. Click **Create Backup**
3. The database is backed up to the configured location
4. Backup file includes timestamp in the filename

Restoring from Backup

1. Select a backup from the list
2. Click **Restore**
3. Confirm the restoration (this will overwrite current data)

5.7 Settings

Configure system-wide settings:

- **Gym Name / Company Name:** Displayed in reports
 - **Currency Symbol:** Used in all financial displays
 - **Logo:** Gym and developer logos
 - **Address / Phone:** Contact information
 - **Password Policy:** Minimum length, complexity requirements
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6. POS Terminal

The Point-of-Sale terminal for product sales.

6.1 Selling Products

Quick Sale

1. Browse products by **category tabs** on the left side
2. Click a product to add it to the cart
3. Adjust **quantity** in the cart if needed
4. Click **Checkout**
5. Select **payment method**:
 - **Cash:** Direct cash payment
 - **Card Balance:** Deduct from player's card balance
6. Confirm the sale
7. Stock is automatically deducted
8. Transaction is logged

Cart Management

- **Remove item:** Click the X button next to an item
- **Change quantity:** Edit the quantity field directly

- **Clear cart:** Remove all items at once
- **Total:** Automatically calculated as items are added/removed

6.2 Card Balance & Top-Up

Checking Balance

1. Search for a player by name in the search bar
2. The player's card balance is displayed
3. Balance is shown in the selected currency

Topping Up Balance

1. Search and select a player
2. Click **Top Up**
3. Enter the top-up amount
4. Confirm the payment
5. Balance is updated immediately
6. An income transaction is recorded

6.3 Barcode Scanner

- Connect a USB barcode scanner to the POS PC
- Scan any product barcode
- The product is automatically found and added to the cart
- If the barcode is not found, an error message is displayed

Today's Sales

- The top bar shows today's total sales count and amount
 - Click **Refresh** to update the figures
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7. License Activation

First-Time Activation

1. On first launch, the **Activation Window** appears
2. Your **Machine ID** is displayed (unique to your PC)
3. Contact the software provider with your Machine ID
4. Enter the **Activation Code** provided
5. Click **Activate**

License Validity

- License is tied to your specific machine (hardware-based)
 - A warning appears 15 days before license expiry
 - Contact your provider to renew the license before it expires
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8. Security Features

AccessControlPro includes enterprise-grade security:

Feature	Description
Password Hashing	PBKDF2 with salting for all user passwords
Data Encryption	AES-256 encryption for sensitive data
DPAPI	Windows-native encryption for device passwords
Brute Force Protection	Account lockout after failed attempts (DB-persistent)

Rate Limiting	100 card operations per minute per device
SQL Injection Prevention	Parameterized queries + table name whitelist
Audit Trail	All operations logged with user, timestamp, details
Concurrency Control	Row versioning prevents lost updates
Permission System	Granular per-feature permissions per user
Photo Encryption	GDPR-compliant photo storage

9. Troubleshooting

Cannot Connect to Database

- Verify SQL Server is running
- Check `appsettings.json` connection string
- Ensure the SQL Server allows TCP/IP connections
- Check firewall settings for port 1433

Device Not Connecting

- Verify the device is powered on and on the same network
- Check the IP address and port in device settings
- Ensure no firewall is blocking the device port
- Try the **Search Network** feature to rediscover devices

Card Not Syncing to Device

- Check that the device is **online** (green indicator)
- Verify the card number is correct
- Ensure the device is selected as a sync target
- Check the rate limiter (max 100 operations per minute per device)

Forgot Admin Password

- Contact your system administrator or software provider
- A password reset requires direct database access

Application Crashes

- Check the `crash_log.txt` file in the application directory
- Check `Logs/session.log` for operation-level details
- Ensure SQL Server is accessible
- Verify sufficient disk space and memory

License Issues

- Ensure the machine hardware has not changed significantly
- Contact the provider with your current Machine ID
- License is hardware-bound and cannot be transferred without a new key